

|   |          |                                                                                                                   |
|---|----------|-------------------------------------------------------------------------------------------------------------------|
|   |          | $F = 200 = 600 \times 0.3$ , giving $F = 600 \text{ N}$<br>Hence power = $Fv = 600 \times 20 = 12\,000 \text{ W}$ |
| 9 | <b>C</b> | Centripetal acceleration<br>$= r\omega^2$<br>$= r(2\pi/T)^2$                                                      |

|    |          |                                                                                               |
|----|----------|-----------------------------------------------------------------------------------------------|
|    |          | Since both $r$ and $T$ are constants, centripetal acceleration does not change over time.     |
| 10 | <b>A</b> | Conservation of energy gives $2400 \times 120 = 0.106 \times 2260000 \times \text{heat loss}$ |